Shahista Tamkeen

Machine Learning Engineer

IL, USA | 331-272-5717| stamkeen0506@gmail.com | LinkedIn | Portfolio | GitHub

PROFESSIONAL SUMMARY

Machine Learning Engineer with 3+ years of professional experience designing, developing, and deploying production-ready AI solutions across fraud detection, predictive analytics, Generative AI, and intelligent automation. Experienced in building scalable machine learning systems, Graph Neural Networks (GNNs), Retrieval-Augmented Generation (RAG) applications, and cloud-native MLOps pipelines. Strong foundation in software engineering, backend development, and distributed data processing, with expertise in delivering reliable, explainable, and production-ready AI solutions using Python, Apache Spark, Kubernetes, and Microsoft Azure.

TECHNICAL SKILLS

Programming Languages: Python, Java, SQL, Bash

Software Engineering: Object-Oriented Programming (OOP), Data Structures & Algorithms, RESTful APIs, FastAPI, Flask, Microservices, Design Patterns, Git, GitHub, CI/CD, Unit Testing, Integration Testing

Machine Learning: Scikit-learn, XGBoost, LightGBM, CatBoost, TensorFlow, PyTorch, PyTorch Geometric (GNNs), Supervised & Unsupervised Learning, Feature Engineering, Hyperparameter Tuning, Model Evaluation, Explainable AI (SHAP, LIME)

Deep Learning: CNNs, RNNs, LSTMs, Transformers, Transfer Learning, Computer Vision

Generative AI & LLMs: LangChain, LangGraph, Azure OpenAI, OpenAI API, Retrieval-Augmented Generation (RAG), AI Agents, Prompt Engineering, Vector Databases, Semantic Search, Llama, Mistral

Data Engineering: Apache Spark, Spark Structured Streaming, Apache Kafka, Snowflake, Pandas, NumPy, ETL Pipelines

MLOps & Deployment: MLflow, Docker, Kubernetes, FastAPI, Model Serving, Model Monitoring, GitHub Actions

Cloud & Databases: Microsoft Azure, AWS, PostgreSQL, SQL Server, Redis, Pinecone

Visualization & Analytics: Power BI, Tableau, Streamlit, Matplotlib

PROFESSIONAL EXPERIENCE

Machine Learning & Generative AI Engineering | AIG, USA *Jan 2026 – Present*

- Developed production-ready machine learning solutions for payment fraud detection using Python, Apache Spark, and XGBoost, improving fraud risk identification by 22% across high-volume financial transaction datasets.
- Built Graph Neural Network (GNN) models using PyTorch Geometric to uncover hidden relationships between merchants, customers, and devices, increasing fraud pattern detection accuracy by 18%.
- Designed and implemented Retrieval-Augmented Generation (RAG) applications using LangChain, LangGraph, Azure OpenAI, and Pinecone, reducing fraud investigation time by approximately 30% through AI-assisted knowledge retrieval.
- Designed and developed RESTful APIs using FastAPI to expose machine learning inference services, enabling scalable integration with enterprise fraud detection platforms.
- Built reusable feature engineering pipelines, experiment tracking workflows, and model evaluation processes using MLflow, improving model reproducibility and deployment readiness.
- Built and maintained containerized machine learning services using Docker and Kubernetes to support scalable deployment, reliable inference, and streamlined release processes.
- Collaborated with senior machine learning engineers, software engineers, data engineers, and fraud analysts to deliver explainable AI solutions while following software engineering best practices, code reviews, Git workflows, and CI/CD pipelines.

Machine Learning Engineer | Persistent Systems, India *Nov 2021 – Jul 2024*

- Developed machine learning models for credit risk assessment, fraud detection, and customer behavior analytics using Python, Scikit-learn, XGBoost, and LightGBM, improving lending risk prediction accuracy by **24%**.
- Designed scalable data preprocessing and feature engineering pipelines using Apache Spark and SQL to process over **5M+** financial records, reducing data preparation time by **30%**.
- Designed and implemented RESTful APIs using FastAPI to integrate machine learning models into enterprise banking applications, enabling real-time risk scoring and improving application scalability.
- Improved model performance through feature engineering, hyperparameter tuning, cross-validation, and Explainable AI (SHAP & LIME), increasing model precision by **21%** while enhancing model interpretability.
- Automated model training, experiment tracking, and deployment workflows using MLflow, Docker, and Apache Airflow, improving deployment consistency and reducing manual operational effort.
- Collaborated with data engineers, software developers, business analysts, and product stakeholders to design, develop, and deploy production-ready machine learning solutions aligned with business requirements.
- Developed interactive dashboards using Power BI and Tableau to visualize model performance, customer segmentation, fraud trends, and business KPIs, enabling data-driven decision-making across cross-functional teams.

EDUCATION

Elmhurst University | Elmhurst, Illinois, USA **Aug 2024 – May 2026**
Master of Science in Computer Science

PROJECTS

Digital Twin Multi-Agent AI Platform

- Designed and developed a production-ready multi-agent AI platform using LangGraph, LangChain, and Azure OpenAI to automate career guidance, financial planning, and personalized learning workflows.
- Implemented Retrieval-Augmented Generation (RAG), semantic search, vector databases, and agent orchestration to deliver contextual, multi-step AI responses with improved accuracy and reliability.
- Built scalable RESTful APIs using FastAPI and integrated PostgreSQL for persistent memory, user management, and secure backend services supporting concurrent AI workflows.
- Technologies: Python, LangGraph, LangChain, Azure OpenAI, FastAPI, PostgreSQL, React, Docker

AI-Powered 3D Point Cloud Quality Assessment System

- Developed an AI-powered computer vision application for automated quality assessment of construction point cloud datasets using Open3D and geometric analysis techniques.
- Implemented point cloud registration, deviation analysis, and defect detection algorithms to improve inspection accuracy and reduce manual quality verification efforts.
- Built an interactive visualization platform using React and Three.js for real-time 3D rendering, engineering model comparison, and defect analysis.
- Technologies: Python, Open3D, Computer Vision, FastAPI, React, Three.js

Brain Age Prediction Using Deep Learning

- Developed a deep learning framework using Convolutional Neural Networks (CNNs) to predict biological brain age from MRI-based neuroimaging datasets.
- Built an end-to-end pipeline including image preprocessing, augmentation, feature extraction, model training, and evaluation to support medical imaging research.
- Published the research in the International Journal of Engineering Research (IJER), demonstrating the application of deep learning techniques in healthcare.
- Technologies: Python, TensorFlow, CNN, Deep Learning, Medical Imaging

Banking Customer Risk & Loan Prediction

- Developed supervised machine learning models to predict customer credit risk and loan approval outcomes using structured financial datasets.
- Performed feature engineering, exploratory data analysis, and model optimization to improve prediction performance and support lending decisions.
- Built interactive dashboards using Streamlit and Power BI to visualize customer segmentation, credit risk, and key business insights for decision-makers.
- Technologies: Python, Scikit-learn, SQL, Streamlit, Power BI

CERTIFICATIONS

- Google Cloud Certification – Google Cloud
- Microsoft Certified: Azure Fundamentals (AZ-900) – Microsoft
- AI & Data Science Certification – IBM
- Java Programming Certification – Oracle
- C Programming Certification (CLA)

PUBLICATIONS

AI for Brain Age Prediction Using Deep Learning
International Journal of Engineering Research (IJER)

- Published research on deep learning-based brain age prediction using MRI neuroimaging datasets.
- Developed a Convolutional Neural Network (CNN) framework for image preprocessing, feature extraction, model training, and biological brain age estimation.